

120 Days of Quant Finance

Week 6: Stochastic Processes & Dynamic Asset Modeling

Day 37: Brownian Motion and Continuous-Time Intuition

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Introduction

Discrete-time random walks provide a useful foundation for modeling asset price movements. However, financial markets evolve continuously, motivating the use of continuous-time stochastic processes. Brownian motion serves as the fundamental building block for such models.

From Random Walks to Continuous Time

A discrete-time random walk can be written as:

$$X_t = \sum_{i=1}^t \varepsilon_i,$$

where ε_i are independent random shocks.

As the time step decreases and the number of steps increases, the properly scaled random walk converges to Brownian motion.

Definition of Brownian Motion

A stochastic process $\{W_t\}_{t \geq 0}$ is called a Brownian motion if it satisfies the following properties:

- $W_0 = 0$
- Independent increments

- Normally distributed increments:

$$W_t - W_s \sim \mathcal{N}(0, t - s), \quad t > s$$

- Continuous sample paths

Variance Growth and Time Scaling

Brownian motion has variance:

$$\text{Var}(W_t) = t.$$

This implies that uncertainty grows linearly with time, while volatility scales with the square root of time.

Drifted Brownian Motion

To incorporate a deterministic trend, Brownian motion is augmented with a drift term:

$$X_t = \mu t + \sigma W_t,$$

where μ represents drift and σ controls volatility.

Relevance to Financial Modeling

Brownian motion forms the basis of many financial models, including geometric Brownian motion and the Black–Scholes option pricing framework. It enables the modeling of continuous-time uncertainty in asset prices.

Conclusion

Brownian motion provides a continuous-time extension of discrete random walks. Understanding its properties is essential for modern financial modeling and forms the foundation of stochastic calculus used in derivative pricing.